British Airways Flight 009

Flight model: Boeing 737
Date: June 24, 1982

From: London, UK
To: Australia

Problem

The plane took off from London. The crew was at the time of the incident boarded the plane at Kaula Lumpur. When passing above Indonesia towards Australia the crew notices smoke inside the cabin. It is worth noting that it was legal to smoke inside the cabin at that period when the incident occurred. But, the smoke was thicker than usual cigarette smoke. Suddenly the windscreen in the cockpit is lit with electric discharges (Analogy to the incident would be plane covered with fireflies), this is called as **Frictional electrification**. The pilots discuss regarding the possibility of **Elmo's fire**, Elmo's fire is a phenomena which is observed when a plane passes through clouds that are charged. In the cabin the smoke becomes more thick with no visible indication of any fire incident in the plane. The pilots were able to smell identical to Sulfur. Then all the four engines of the 737 were seen to throw a huge fire tail behind the engines (as long as 20 feet). Then the engines number four gets powered down followed by number two then the next two engines fail at the same time. The plane declares an emergency but the communication system of the plane fails to establish a proper communication with the ATC. It is worth noting that the 737 without is engine power can glide over 15 KM for every 1 KM drop in altitude. Ideally when all four engines are turned off, the auto pilot must be disabled automatically but, in the case of BA009 the auto pilots were turned on even after the four engines lost its power. The crew turns back to Haleem, the nearest airport in Indonesia. The ATC but is not able to find the plane on its radar. Each of two pilots were able to find two different altitude reading from their Vertical altimeter. The pilots tries to inject more air into the engine by diving the nose of the plane up and down. The pressure inside the cabin started to increase above safer threshold. One of the pilots oxygen mask was not working and the crew decided to dive into a safer altitude so that the crew can breath without the need for oxygen mask. After multiple attempts engine four was back online followed by engine three then other two were back online immediately. When the crew tries to increase the altitude the lights starts appearing again. Then one of the engines losses power again. When the pilots descended the pilots notices that the flickering lights were now not visible but the windscreen was now was badly damaged (Analogy to that would be a glass covered with mist), this significantly reduced the pilots visibility. The only part of the windscreen that was not affected was the boarders about 2 inches remained unaffected. Also, the equipment that help pilots descend the flight at a proper angles (Slide slope) were also found faulty but the equipment that gives the center line of the plane was in service. The crew finally manages to land the flight safely.

Investigation

In the investigation, it was found that the engines, the plane body were badly sanded such that the paint coated on the metal was stripped off. The engines were chocked with fine dust, which in further analysis were found to be volcanic ash. It was found that when the was flying above Indonesia, there was a volcanic eruption in the same place. These ash are fine particles of minerals unlike sand which is of large particles. The particles over the windscreen would have scratched the top layer thereby reducing the visibility this phenomena is called **abrasion**. The particles when hit by the moving plane caused to them to be charge which resulted in the light near the windscreen and disturbance in the communication system. The thick smoke in the cabin was also due to some of ash particles that got suck into the cabin and a smell of Sulfur presence also felt. When the same as is passed through the engine it melts and sticks itself to the engine and starts to chock the engine. The fire in the tail of the engine is when the combustion incomplete here, more fuel less oxygen. This phenomena is called **back fire**. When the plane descended to a low altitude the molten ash cooled down and stared to break of which allowed air to combine with to fuel to restart the engines.

Changes that were brought after the incident

The report advises the pilots to be trained to identify and to respond when the pass through an ash cloud. Also it is learned that dues to absences of water in these ashes they do not appear on the weather radar located in the cockpit. This was resolved by better communication between the Aviation department with the geologist who study volcanic activity.

Reference

1. "The Impossible" That Happened To British Airways Flight 009 — FULL EPISODE — Mayday: Air Disaster
   `https://www.youtube.com/watch?v=riBqZLpTv0o`